GRAPH COURSES ONLINE WORKSHOP_4

1. Summary of evidence

We looked at 19 experiments (from 14 unique studies/papers) in which people were asked to quit or cut down on social media. For positive mood, 3 found an improvement (Allcott et al., 2020; Arceneaux et al., 2024 ,Faulhaber et al., 2023), 11 found no clear effect (Arceneaux et al., 2024 [Deactivation + Information]; Asimovic et al., 2021; Hartanto et al., 2025 [Full Abstinence]; Hartanto et al., 2025 [Partial Abstinence]; Maerevoet et al., 2025; Przybylski et al., 2021 [Experiments 1, 2, & 3]; Vally et al., 2019; Van Wezel et al., 2021; Walsh et al., 2024 [Digital Diet]), and 4 found that mood got worse (Hanley et al., 2019; Vanman et al., 2018; Wadsley et al., 2023; Walsh et al., 2024 [Social Diet]).

The breaks were mostly short, lasting between 1 and 75 days (with all but one lasting between 1 and 28 days), and most participants were university students (13 out of 19 experiments). Studies that asked people to quit completely found mixed results: 2 showed an improvement, 7 found no clear effect, and 4 found that mood got worse. Meanwhile, studies that asked people to cut down (limit usage) found mostly no clear effect (4 out of 6 experiments), with only 1 finding an improvement and 1 not measuring positive mood. The largest study, Allcott et al. (2020), with 1,661 participants, found an improvement in positive mood. Overall, these studies suggest that the results are mixed: while a few longer, larger-scale interventions improved positive mood, the majority of shorter abstinence or time-limit experiments showed no clear benefit or even a temporary decrease in positive mood.

2. Comparison with the meta-analysis.

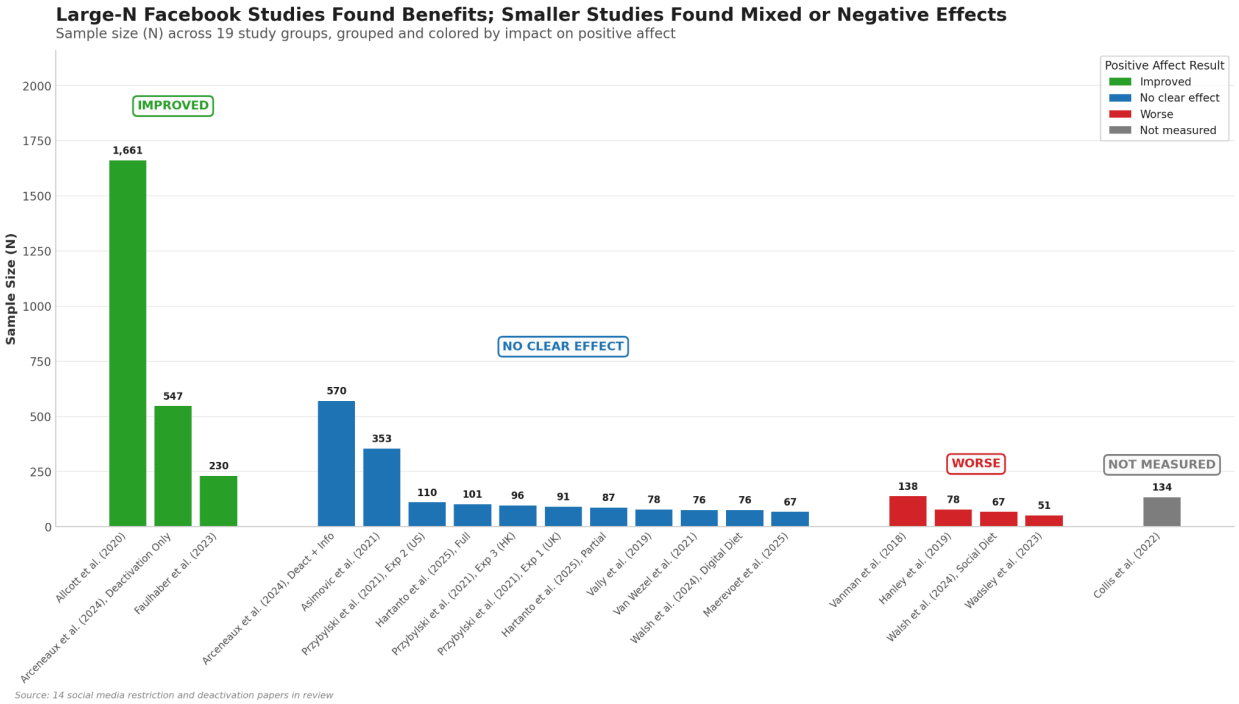
Whereas the table uses categorical "vote counting" ("Improved," "No clear effect," or "Worse") on a hand-picked set of studies, Lemahieu et al. (2025) synthesize systematically selected experiments into a precision-weighted, continuous pooled effect size ($Hedges'g = 0.03$) with 95% confidence intervals ($[-0.11, 0.16]$). This meta-analytic estimate quantifies both the exact magnitude and statistical uncertainty of each study, revealing that overall positive mood changes from complete abstinence are negligible ($p = 0.69$) despite substantial between-study heterogeneity ($I^2 = 61\%$, prediction interval $[-0.42, 0.47]$).

Furthermore, standardized effect sizes resolve discrepancies hidden by simple labels—for example, showing that most individual confidence intervals cross zero and that Vanman et al. (2018) had a positive point estimate for positive affect ($g = 0.34, [-0.02, 0.69]$) rather than the "Worse" label in the table.

3. Use of AI

I used Gemini Notebook (Google) to extract data and summarize findings from the 14 selected studies. I checked them against the source articles. I wrote the text myself.

4. Figure



5. Appendix: Extraction table.

Study	Design	Sample description	Intervention	Positive affect result
Allcott et al. (2020)	RCT	1,661; adult Facebook users, USA	Quit; Facebook; 28	Improved
Arceneaux et al. (2024), Deactivation Only	RCT	547; adult Facebook users, France	Quit; Facebook; 28	Improved

Arceneaux et al. (2024), Deactivation + Information	RCT	570; adult Facebook users, France	Quit; Facebook; 28	No clear effect
Asimovic et al. (2021)	RCT	353; Facebook users, Bosnia and Herzegovina	Quit; Facebook; 7	No clear effect
Collis et al. (2022)	RCT	134; university students, Europe	Limit; Facebook, Instagram, Snapchat, 10 min/day; 75	Not measured
Faulhaber et al. (2023)	RCT	230; university students, USA	Limit; all social media, 30 min/day; 14	Improved
Hanley et al. (2019)	RCT	78; adults, UK, USA, Australia	Quit; Facebook and Instagram; 7	Worse
Hartanto et al. (2025), Full Abstinence	RCT	101; university students, Singapore	Quit; social media (6 apps); 7	No clear effect

Hartanto et al. (2025), Partial Abstinence	RCT	87; university students, Singapore	Limit; social media (6 apps), 10 min/app/day; 7	No clear effect
Maerevoet et al. (2025)	RCT	67; university students, Belgium	Limit; social media, 30 min/day; 14	No clear effect
Przybylski et al. (2021), Experiment 1 (UK)	Within-person	91; university students, UK	Quit; social media; 1	No clear effect
Przybylski et al. (2021), Experiment 2 (US)	Within-person	110; university students, USA	Quit; social media; 1	No clear effect
Przybylski et al. (2021), Experiment 3 (Hong Kong)	Within-person	96; university students, Hong Kong	Quit; social media; 1	No clear effect
Vally et al. (2019)	RCT	78; university students, UAE	Quit; social media; 7	No clear effect
Vanman et al. (2018)	RCT	138; active Facebook users, Australia	Quit; Facebook; 5	Worse

Van Wezel et al. (2021)	RCT	76; university students, Netherlands	Limit; Facebook, Instagram, Snapchat, YouTube, 50% reduction; 7	No clear effect
Wadsley et al. (2023)	Within-person	51; university students, UK	Quit; social media; 7	Worse
Walsh et al. (2024), Digital Diet	RCT	76; university students, USA	Limit; smartphone digital media; 8	No clear effect
Walsh et al. (2024), Social Diet	RCT	67; university students, USA	Quit; social media; 8	Worse